Comparative and Trend Analysis of Agricultural Export Values (2021-2023)

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RESEARCH PROBLEM

To explore if the agricultural export patterns vary significantly across regions and over time.

Understanding these variations is crucial for trade policy, economic planning, and market forecasting. This study investigates:

1. Differences in export values between Gulf and non-Gulf countries
2. Temporal trends across years and seasons
3. Item-specific growth/decline patterns
4. Country clustering based on export profiles

RESEARCH QUESTIONS

1. Is there a significant difference in mean export VALUE between Gulf and Non-Gulf countries? (Wilcoxon Rank-Sum Test)
2. Does export VALUE vary significantly across years (2021-2023)? (Kruskal–Wallis Test)
3. Does export VALUE of Dates vary across seasons in Gulf countries? (Kruskal-Wallis Test)
4. Which items show significant growth/decline trends over 2021-2023? (Spearman's Rank Correlation)
5. Can the export profiles of countries be effectively reduced into fewer dimensions using Principal Component Analysis (PCA), and what do these components represent? (Principal Component Analysis (PCA))
6. How can countries be clustered based on their export profiles? (K-means Clustering on PCA scores with elbow/silhouette validation)

SIGNIFICANCE OF THE STUDY

1. Provides empirical evidence for regional export disparities
2. Identifies seasonal patterns for strategic planning
3. Highlights trending items for investment/trade decisions
4. Offers clustering methodology for market segmentation
5. Supports data-driven agricultural trade policies

METHODS & ASSUMPTIONS

```r
# LOAD REQUIRED PACKAGES

#install.packages("caret")
library(readxl)
library(car)
library(ggplot2)
library(dplyr)
library(scales)
library(boot)
library(tidyverse)
library(ggpubr)
library(kableExtra)
library(FactoMineR)
library(factoextra)
library(corrplot)
library(cluster)
library(NbClust)
library(ggrepel)
library(RColorBrewer)
library(caret)
library(pROC)
library(plotly)
```

```r
# DATA LOADING
my_data <- read_excel("Data_Export.xlsx")
```

Statistical Assumptions

1. **Normality**: Data should be approximately normally distributed (Shapiro-Wilk test, Histogram and QQ plots).
2. **Independence**: Observations should be independent of each other.
3. **Homogeneity of Variance**: Equal variances across groups (Levene's test, and Box-plot).

Part I: Inferential Analysis

Classical Inferential Procedures:

Q1. Is there a significant difference in mean export VALUE between Gulf and Non-Gulf countries?

- Method: Wilcoxon Rank-Sum Test (non-parametric alternative to t-test)
- Mathematical formulation:
  - $W = sum(R_i)$ where R_i are ranks
  - Under H0: W ~ Normal(n1(n1+n2+1)/2, sqrt(n1*n2*(n1+n2+1)/12))

```r
# CREATE GULF/NON-GULF CLASSIFICATION
# Define Gulf countries and create a new GROUP variable
# Countries are classified as "Gulf" if they match the predefined list
# All other countries are classified as "NonGulf"

gulf_countries <- c("Oman", "UAE", "United Arab Emirates",
                    "Saudi Arabia", "KSA", "Kuwait",
                    "Qatar", "Bahrain")

my_data$GROUP <- ifelse(my_data$COUNTRY %in% gulf_countries,
                        "Gulf", "NonGulf")

my_data$GROUP <- factor(my_data$GROUP)
```

```r
# SHAPIRO-WILK TEST FOR NORMALITY
# Shapiro-Wilk test for Gulf countries
shapiro_gulf <- my_data %>%
  filter(GROUP == "Gulf") %>%
  pull(VALUE) %>%
  shapiro.test()

# Shapiro-Wilk test for Non-Gulf countries
shapiro_nongulf <- my_data %>%
  filter(GROUP == "NonGulf") %>%
  pull(VALUE) %>%
  shapiro.test()

cat("\nGulf Countries:\n")
```

```
##
## Gulf Countries:
```

```r
print(shapiro_gulf)
```

```
##
##  Shapiro-Wilk normality test
##
## data:  .
## W = 0.10063, p-value < 2.2e-16
```

```r
cat("\nNon-Gulf Countries:\n")
```

```
##
## Non-Gulf Countries:
```

```r
print(shapiro_nongulf)
```

```
##
##  Shapiro-Wilk normality test
##
## data:  .
## W = 0.30123, p-value < 2.2e-16
```

```r
# LEVENE'S TEST FOR HOMOGENEITY OF VARIANCE
# Levene's test for Gulf vs. Non-Gulf groups
# Test with median
levene_gulf <- leveneTest(VALUE ~ GROUP, data = my_data, center = median)
cat("\nLevene's Test (Gulf vs. Non-Gulf) - Median:\n")
```

```
##
## Levene's Test (Gulf vs. Non-Gulf) - Median:
```

```r
print(levene_gulf)
```

```
## Levene's Test for Homogeneity of Variance (center = median)
##         Df F value    Pr(>F)
## group    1  129.66 < 2.2e-16 ***
##       5682
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```r
# Test with mean
levene_gulf_mean <- leveneTest(VALUE ~ GROUP, data = my_data, center = mean)
cat("\nLevene's Test (Gulf vs. Non-Gulf) - Mean:\n")
```

```
##
## Levene's Test (Gulf vs. Non-Gulf) - Mean:
```

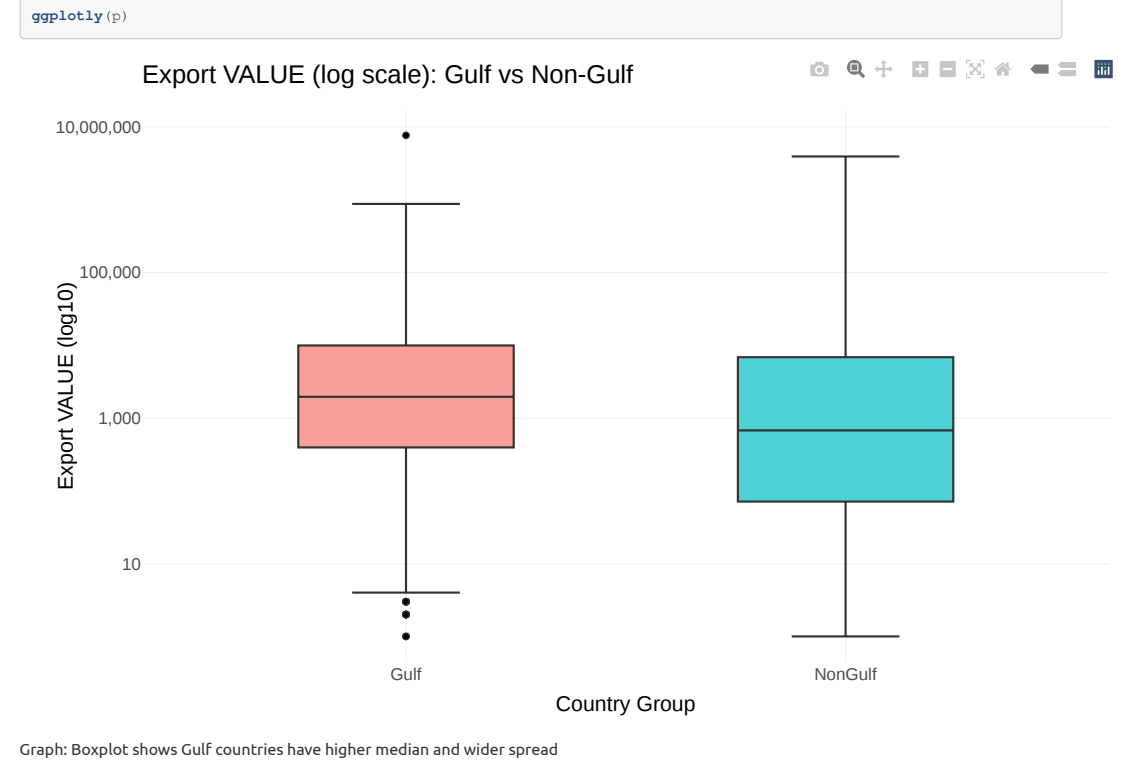
```r
print(levene_gulf_mean)
```

```
## Levene's Test for Homogeneity of Variance (center = mean)
##         Df F value    Pr(>F)
## group    1  480.26 < 2.2e-16 ***
##       5682
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

For normality: Gulf countries: Data is NOT normally distributed (p = 1.980565e-90) < 0.05 Non-Gulf countries: Data is NOT normally distributed (p = 2.298397e-55) < 0.05

For homogeneity of variance: Variances are NOT equal between Gulf and Non-Gulf groups (p = 1.017163e-29) < 0.05 Assumption of homogeneity of variance is VIOLATED

```r
# VISUALIZATION: BOXPLOT COMPARING GULF VS NON-GULF COUNTRIES
p <- ggplot(my_data, aes(x = GROUP, y = VALUE, fill = GROUP)) +
  geom_boxplot(alpha = 0.7) +
  scale_y_log10(labels = scales::comma) +
  labs(title = "Export VALUE (log scale): Gulf vs Non-Gulf",
       x = "Country Group",
       y = "Export VALUE (log10)") +
  theme_minimal(base_size = 14) +
  theme(legend.position = "none")

ggplotly(p)
```



Graph: Boxplot shows Gulf countries have higher median and wider spread

```r
# WILCOXON RANK-SUM TEST (NON-PARAMETRIC T-TEST)
wilcox_result <- wilcox.test(VALUE ~ GROUP,
                             data = my_data,
                             exact = FALSE)   # Use normal approx for large samples

wilcox_result
```

```
##
##  Wilcoxon rank sum test with continuity correction
##
## data:  VALUE by GROUP
## W = 3233494, p-value < 2.2e-16
## alternative hypothesis: true location shift is not equal to 0
```

Result: W = 3,233,494, p < 2.2e-16 Interpretation: Strong evidence that Gulf countries have higher export values

```r
#MEDIANS
my_data %>%
  group_by(GROUP) %>%
  summarise(median_VALUE = median(VALUE, na.rm = TRUE),
            n = n())
```

```
## # A tibble: 2 × 3
##   GROUP   median_VALUE     n
##   <fct>          <dbl> <int>
## 1 Gulf            1961  4474
## 2 NonGulf          677  1211
```

Median Gulf: 1961 vs Non-Gulf: 677 Therefore, we conclude that Gulf countries exhibit significantly higher export VALUE than Non-Gulf countries.

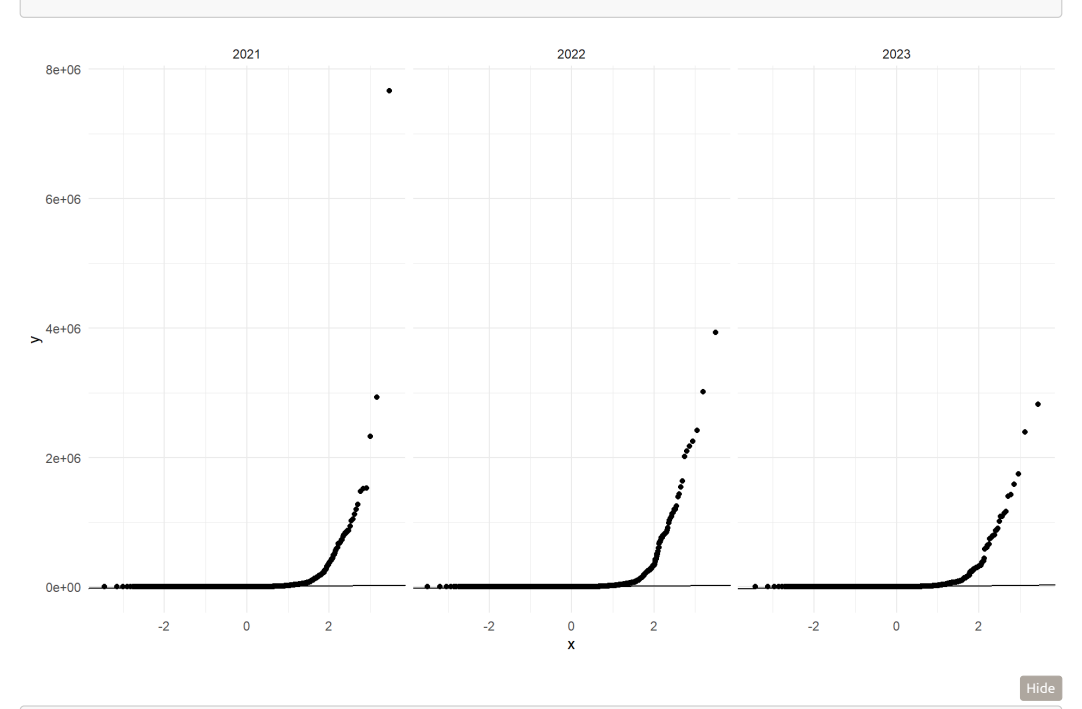
Q2. Does export VALUE vary significantly across years (2021-2023)?

- Method: Kruskal-Wallis Test (non-parametric alternative to ANOVA)
- Mathematical formulation:
  - $H = [12/(N(N+1))] * \Sigma(T_j^2/n_j) - 3(N+1)$
  - where T_j = sum of ranks in group j

```r
# Convert YEAR to factor for categorical analysis
my_data$YEAR <- as.factor(my_data$YEAR)

# SHAPIRO-WILK TEST FOR NORMALITY
shapiro_by_year <- my_data %>%
  group_by(YEAR) %>%
  summarise(
    W = shapiro.test(VALUE)$statistic,
    p_value = shapiro.test(VALUE)$p.value,
    n = n(),
    .groups = 'drop'
  )
print(shapiro_by_year)
```

```
## # A tibble: 3 × 4
##   YEAR      W  p_value     n
##   <fct> <dbl>    <dbl> <int>
## 1 2021  0.122 8.95e-69  1877
## 2 2022  0.178 2.46e-70  2129
## 3 2023  0.209 2.59e-64  1679
```

```r
# QQ plots
ggplot(my_data, aes(sample = VALUE)) +
  stat_qq() +
  stat_qq_line() +
  facet_wrap(~ YEAR) +
  theme_minimal()
```



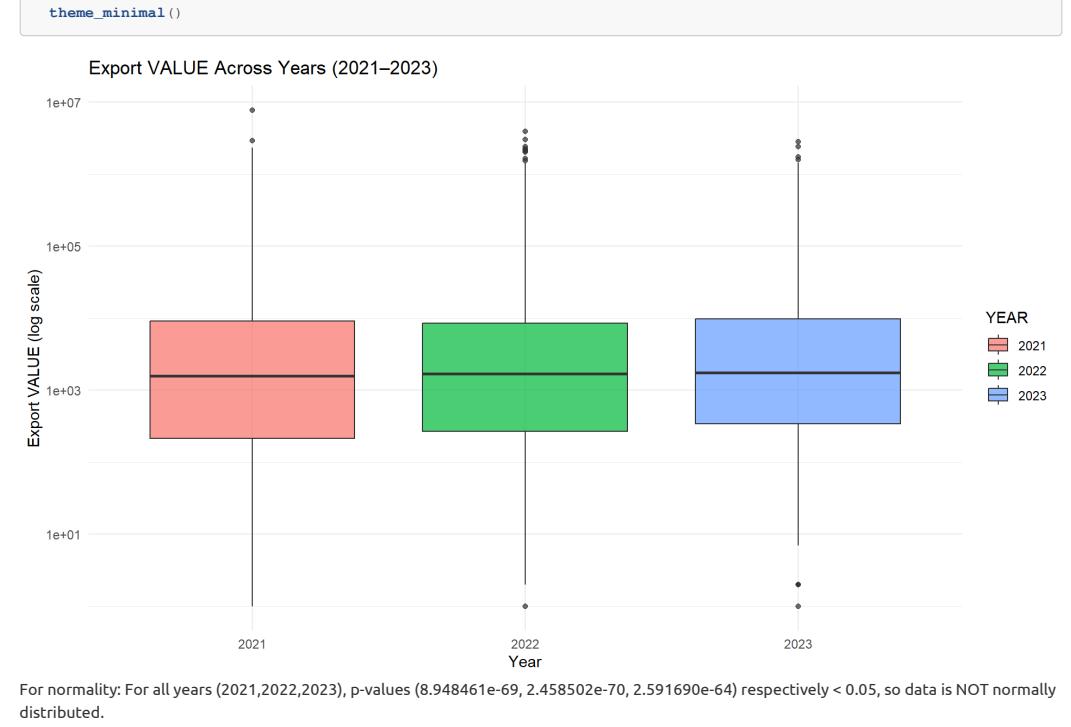
```r
# LEVENE'S TEST FOR HOMOGENEITY OF VARIANCE
levene_years <- leveneTest(VALUE ~ YEAR, data = my_data, center = median)
cat("\nLevene's Test (Across Years) - Median:\n")
```

```
##
## Levene's Test (Across Years) - Median:
```

```r
print(levene_years)
```

```
## Levene's Test for Homogeneity of Variance (center = median)
##         Df F value Pr(>F)
## group    2  0.1792 0.8359
##       5681
```

```r
# Boxplot
ggplot(my_data, aes(x = YEAR, y = VALUE, fill = YEAR)) +
  geom_boxplot(alpha = 0.7) +
  scale_y_log10() +
  labs(title = "Export VALUE Across Years (2021–2023)",
       y = "Export VALUE (log scale)",
       x = "Year") +
  theme_minimal()
```



For normality: For all years (2021,2022,2023), p-values (8.948461e-69, 2.458502e-70, 2.591690e-64) respectively < 0.05, so data is NOT normally distributed.

For homogeneity of variance: Variances are equal across years. ANOVA assumption of homogeneity of variance is SATISFIED

```r
# Kruskal-Wallis Test
kruskal_result <- kruskal.test(VALUE ~ YEAR, data = my_data)
kruskal_result
```

```
##
##  Kruskal-Wallis rank sum test
##
## data:  VALUE by YEAR
## Kruskal-Wallis chi-squared = 6.3714, df = 2, p-value = 0.04135
```

Result: Kruskal-Wallis χ^2 = 6.3714, p = 0.04135 Since the p-value (0.04135) < 0.05, there is significant difference in export values across years.

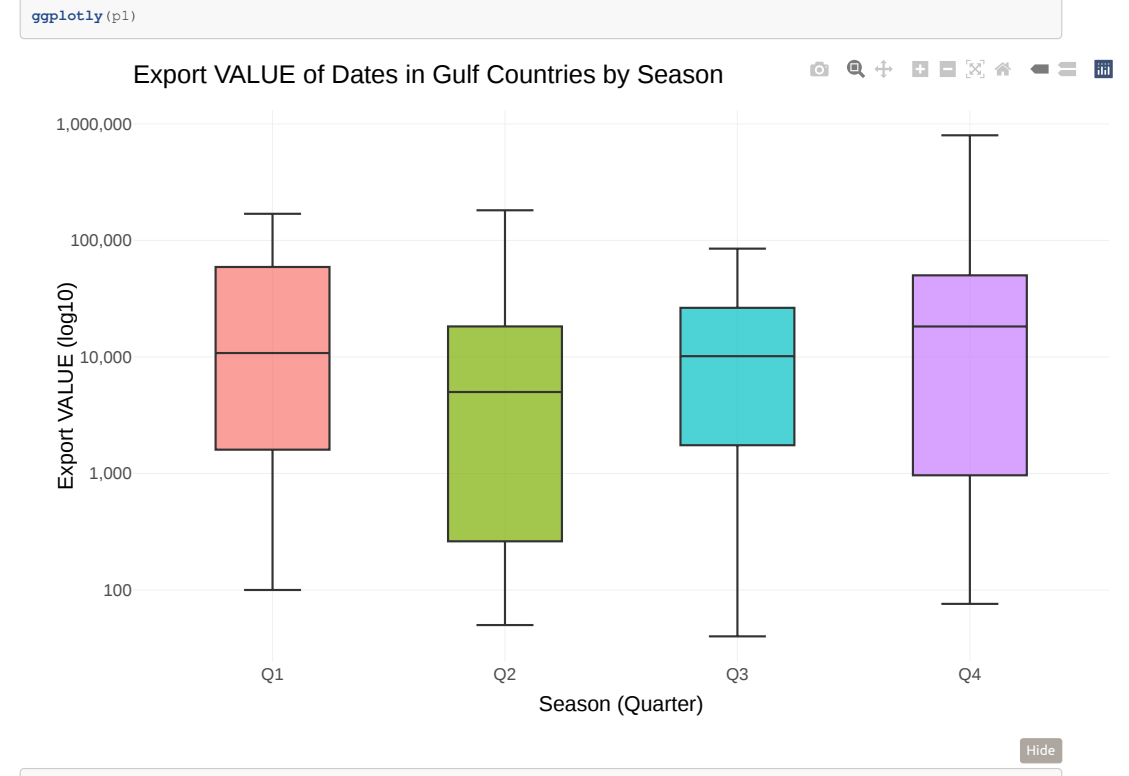
Q3. Does export VALUE of Dates vary across seasons in Gulf countries?

- Method: Kruskal-Wallis Test (non-parametric alternative to ANOVA)
- Mathematical formulation:
  - $H = [12/(N(N+1))] * \Sigma(T_j^2/n_j) - 3(N+1)$
  - where:
    - N = total number of observations
    - T_j = sum of ranks in season j
    - n_j = number of observations in season j

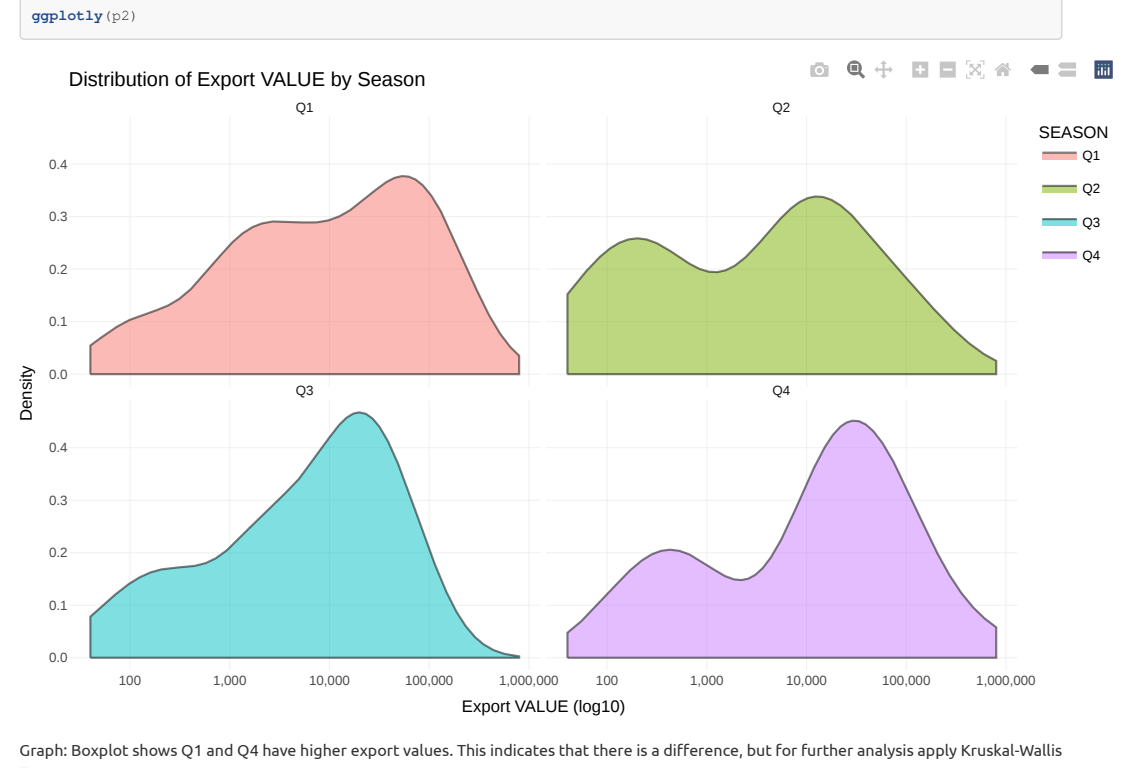
```r
# Filter data to include only:
# 1. Gulf countries
# 2. Date-related items (case-insensitive match)
dates_data <- my_data %>%
  filter(COUNTRY %in% gulf_countries &
           grepl("Date", ITEM, ignore.case = TRUE)) %>%
  mutate(
    # Standardize country names
    COUNTRY = case_when(
      COUNTRY %in% c("UAE", "United Arab Emirates") ~ "UAE",
      COUNTRY == "KSA" ~ "Saudi Arabia",
      TRUE ~ COUNTRY
    )
  )

# Create SEASON variable based on MONTH
dates_data <- dates_data %>%
  mutate(
    SEASON = case_when(
      MONTH %in% 1:3 ~ "Q1",
      MONTH %in% 4:6 ~ "Q2",
      MONTH %in% 7:9 ~ "Q3",
      MONTH %in% 10:12 ~ "Q4",
      TRUE ~ NA_character_
    ),
    SEASON = factor(SEASON, levels = c("Q1", "Q2", "Q3", "Q4"))
  )

# Remove any rows with missing season
dates_data <- dates_data %>% filter(!is.na(SEASON))

# Check sample sizes per season
dates_data %>%
  group_by(SEASON) %>%
  summarise(
    n = n(),
    mean_VALUE = mean(VALUE, na.rm = TRUE),
    median_VALUE = median(VALUE, na.rm = TRUE),
    sd_VALUE = sd(VALUE, na.rm = TRUE)
  )
```

```
## # A tibble: 4 × 5
##   SEASON     n mean_VALUE median_VALUE sd_VALUE
##   <fct>  <int>      <dbl>        <dbl>    <dbl>
## 1 Q1        41     38513.        10800   50513.
## 2 Q2        43     22398.         5000   42741.
## 3 Q3        36     17633.        10155   22000.
## 4 Q4        41     56244.        18255  132482.
```

```r
# Exploratory Visualization
# Boxplot of VALUE by SEASON
p1 <- ggplot(dates_data, aes(x = SEASON, y = VALUE, fill = SEASON)) +
  geom_boxplot(alpha = 0.7) +
  scale_y_log10(labels = scales::comma) +
  labs(title = "Export VALUE of Dates in Gulf Countries by Season",
       subtitle = "Log scale",
       x = "Season (Quarter)",
       y = "Export VALUE (log10)") +
  theme_minimal(base_size = 14) +
  theme(legend.position = "none")

ggplotly(p1)
```



```r
# Density plot for each season
p2 <- ggplot(dates_data, aes(x = VALUE, fill = SEASON)) +
  geom_density(alpha = 0.5) +
  scale_x_log10(labels = scales::comma) +
  facet_wrap(~ SEASON, ncol = 2) +
  labs(title = "Distribution of Export VALUE by Season",
       x = "Export VALUE (log10)",
       y = "Density") +
  theme_minimal()

ggplotly(p2)
```



Graph: Boxplot shows Q1 and Q4 have higher export values. This indicates that there is a difference, but for further analysis apply Kruskal-Wallis Test.

```r
# Kruskal-Wallis Test (non-parametric alternative)
kruskal_result <- kruskal.test(VALUE ~ SEASON, data = dates_data)
print("Kruskal-Wallis Test Results:")
```

```
## [1] "Kruskal-Wallis Test Results:"
```

```r
print(kruskal_result)
```

```
##
##  Kruskal-Wallis rank sum test
##
## data:  VALUE by SEASON
## Kruskal-Wallis chi-squared = 7.392, df = 3, p-value = 0.0604
```

```r
# Extract chi-squared and p-value for reporting
chi_sq <- kruskal_result$statistic
p_value_kw <- kruskal_result$p.value

# Check with different alpha level (e.g., α=0.10)
alpha <- 0.10
if (kruskal_result$p.value < alpha) {
  print(paste("Significant at α =", alpha))
} else {
  print(paste("Not significant at α =", alpha))
}
```

```
## [1] "Significant at α = 0.1"
```

```r
# Look at effect size
epsilon_squared <- function(kruskal_result, n_total) {
  h <- kruskal_result$statistic
  k <- length(unique(dates_data$SEASON))
  return((h - (k - 1)) / (n_total - 1))
}

eps_sq <- epsilon_squared(kruskal_result, nrow(dates_data))
print(paste("Effect size (epsilon-squared):", round(eps_sq, 3)))
```

```
## [1] "Effect size (epsilon-squared): 0.027"
```

Result: Kruskal-Wallis χ^2 = 7.392, p = 0.0604 Due to the kewens in the graphs and kruskal test value 0.0604, we can expand the significance level to 10% to capture the difference in the the seasons.

Correlation and Association:

Q4. Which items show significant growth/decline trends over 2021-2023?

- Method: Spearman's Rank Correlation
- Mathematical formulation:
  - $\rho = 1 - [6\Sigma d_i^2/(n(n^2-1))]$
  - where d_i = difference in ranks

```r
# Ensure YEAR is numeric
my_data$YEAR <- as.numeric(as.character(my_data$YEAR))
# 1. Filter items with sufficient data
valid_items <- my_data %>%
  group_by(ITEM) %>%
  summarise(
    n = n(),
    n_years = n_distinct(YEAR),
    .groups = 'drop'
  ) %>%
  filter(n >= 10, n_years >= 3)

analysis_data <- my_data %>%
  filter(ITEM %in% valid_items$ITEM) %>%
  filter(!is.na(YEAR), !is.na(VALUE))  # Remove missing values

cat("Analyzing", nrow(valid_items), "items with sufficient data\n")
```

```
## Analyzing 110 items with sufficient data
```

```r
# Calculate Spearman correlation
year_cor <- analysis_data %>%
  group_by(ITEM) %>%
  summarise(
    n = n(),
    n_years = n_distinct(YEAR),
    rho = cor(VALUE, YEAR, method = "spearman", use = "complete.obs"),
    .groups = 'drop'
  ) %>%
  filter(!is.na(rho)) %>%
  # Add p-values separately to avoid errors in summarise
  mutate(
    p_value = map_dbl(ITEM, ~{
      item_data <- analysis_data %>% filter(ITEM == .x)
      if(nrow(item_data) >= 3) {
        cor_test <- cor.test(item_data$VALUE, item_data$YEAR,
                             method = "spearman", exact = FALSE)
        return(cor_test$p.value)
      } else {
        return(NA)
      }
    })
  ) %>%
  mutate(
    trend = case_when(
      rho > 0.3 & p_value < 0.05 ~ "Strong Growth",
      rho > 0 & p_value < 0.05 ~ "Weak Growth",
      rho < -0.3 & p_value < 0.05 ~ "Strong Decline",
      rho < 0 & p_value < 0.05 ~ "Weak Decline",
      TRUE ~ "No Clear Trend"
    ),
    significance = ifelse(p_value < 0.05, "*", "")
  ) %>%
  arrange(desc(abs(rho)))
```

```r
# Summary statistics
#TREND SUMMARY
trend_summary <- year_cor %>%
  group_by(trend) %>%
  summarise(count = n(), percent = round(100*n()/nrow(year_cor), 1), .groups = 'drop')
print(trend_summary, n = Inf)
```

```
## # A tibble: 5 × 3
##   trend          count percent
##   <chr>          <int>   <dbl>
## 1 No Clear Trend    92    83.6
## 2 Strong Decline     5     4.5
## 3 Strong Growth     10     9.1
## 4 Weak Decline       2     1.8
## 5 Weak Growth        1     0.9
```

```r
# TOP 10 GROWING ITEMS
year_cor %>%
  filter(rho > 0, p_value < 0.05) %>%
  arrange(desc(rho)) %>%
  slice_head(n = 10) %>%
  select(ITEM, rho, p_value, n, n_years, trend) %>%
  print(n = 10)
```

```
## # A tibble: 10 × 6
##    ITEM        rho   p_value     n n_years trend
##    <chr>     <dbl>     <dbl> <int>   <int> <chr>
##  1 Forage    0.612 0.000697     27       3 Strong Growth
##  2 Peas      0.574 0.00526      22       3 Strong Growth
##  3 Rice      0.483 0.0108       27       3 Strong Growth
##  4 Lentils   0.442 0.0146       30       3 Strong Growth
##  5 Wheat     0.433 0.00336      44       3 Strong Growth
##  6 Marrow    0.402 0.0166       35       3 Strong Growth
##  7 Cabbage   0.400 0.0000452    98       3 Strong Growth
##  8 Cucumbers 0.360 0.000416     92       3 Strong Growth
##  9 Oats      0.355 0.0266       39       3 Strong Growth
## 10 Tea       0.342 0.0309       40       3 Strong Growth
```

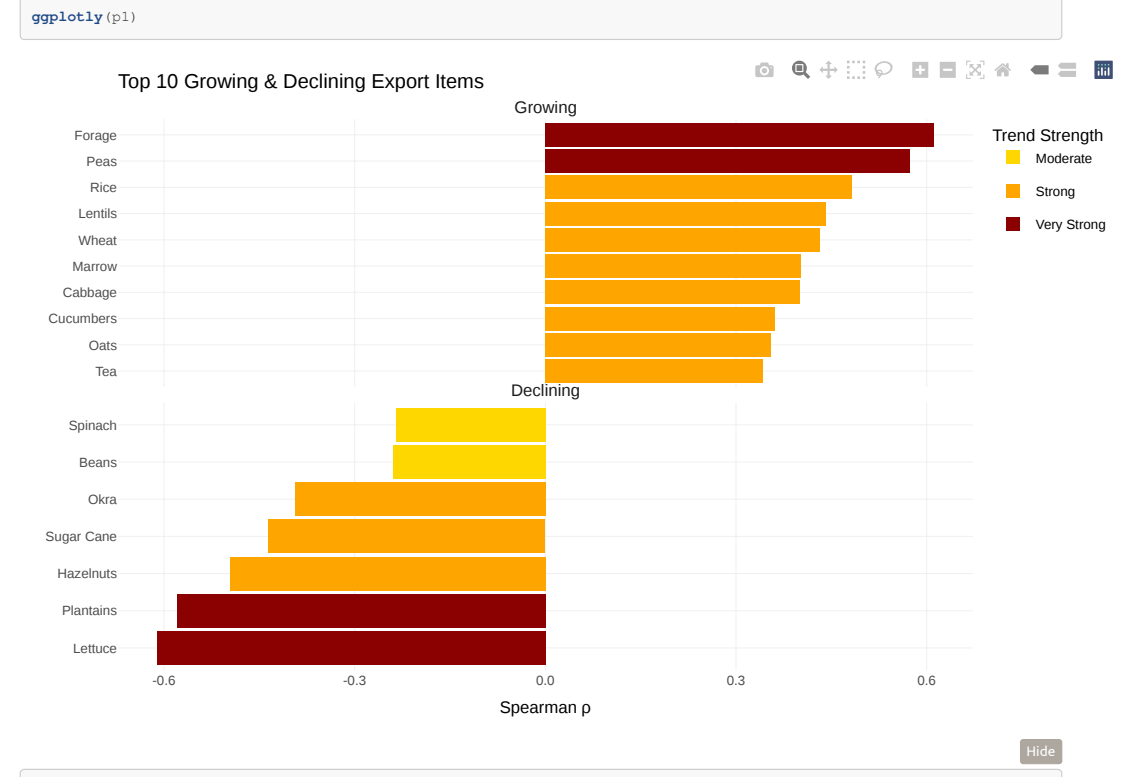
```r
# TOP 10 DECLINING ITEMS
year_cor %>%
  filter(rho < 0, p_value < 0.05) %>%
  arrange(rho) %>%
  slice_head(n = 10) %>%
  select(ITEM, rho, p_value, n, n_years, trend) %>%
  print(n = 10)
```

```
## # A tibble: 7 × 6
##   ITEM          rho    p_value     n n_years trend
##   <chr>       <dbl>      <dbl> <int>   <int> <chr>
## 1 Lettuce    -0.611 0.00000510    47       3 Strong Decline
## 2 Plantains  -0.580 0.0000370     44       3 Strong Decline
## 3 Hazelnuts  -0.496 0.0309        19       3 Strong Decline
## 4 Sugar Cane -0.436 0.0112        33       3 Strong Decline
## 5 Okra       -0.394 0.00153       62       3 Strong Decline
## 6 Beans      -0.240 0.0212        92       3 Weak Decline
## 7 Spinach    -0.235 0.0107       117       3 Weak Decline
```

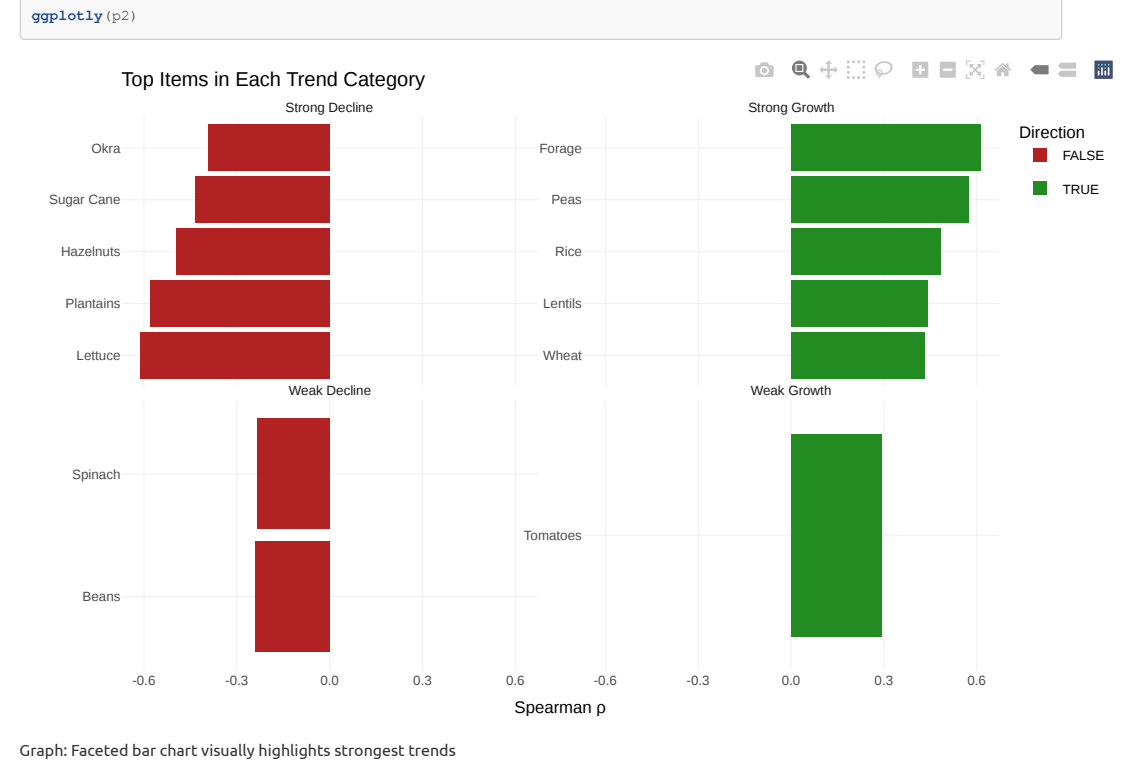
Top Growing Items: Forage (ρ=0.612), Peas (ρ=0.574), Rice (ρ=0.483) Top Declining Items: Lettuce (ρ=-0.611), Plantains (ρ=-0.580)

```r
# Faceted visualization for top 10 growing/declining items

# Get top 10 growing and top 10 declining items
top_growing <- year_cor %>%
  filter(rho > 0, p_value < 0.05) %>%
  arrange(desc(rho)) %>%
  slice_head(n = 10) %>%
  mutate(direction = "Growing")

top_declining <- year_cor %>%
  filter(rho < 0, p_value < 0.05) %>%
  arrange(rho) %>%
  slice_head(n = 10) %>%
  mutate(direction = "Declining")

# Combine and prepare for plotting
top_items <- bind_rows(top_growing, top_declining) %>%
  mutate(
    direction = factor(direction, levels = c("Growing", "Declining")),
    trend_strength = case_when(
      abs(rho) >= 0.5 ~ "Very Strong",
      abs(rho) >= 0.3 ~ "Strong",
      TRUE ~ "Moderate"
    ),
    ITEM_short = str_trunc(ITEM, width = 30)  # Truncate long item names
  )
```

```r
# Faceted bar chart
p1 <- ggplot(top_items, aes(x = reorder(ITEM_short, rho), y = rho,
                            fill = trend_strength)) +
  geom_col() +
  facet_wrap(~ direction, scales = "free_y", ncol = 1) +
  coord_flip() +
  scale_fill_manual(values = c("Very Strong" = "darkred",
                               "Strong" = "orange",
                               "Moderate" = "gold")) +
  labs(title = "Top 10 Growing & Declining Export Items",
       subtitle = "Spearman ρ with Year | Sorted by trend strength",
       x = "",
       y = "Spearman ρ",
       fill = "Trend Strength") +
  theme_minimal() +
  theme(legend.position = "bottom",
        strip.text = element_text(size = 11, face = "bold"))

ggplotly(p1)
```



```r
# Faceted by trend category
year_cor_faceted <- year_cor %>%
  mutate(trend_category = factor(trend,
                                 levels = c("Strong Growth", "Weak Growth",
                                            "No Clear Trend",
                                            "Weak Decline", "Strong Decline")))

# Get top 5 from each significant category
top_by_category <- year_cor_faceted %>%
  filter(trend %in% c("Strong Growth", "Weak Growth", "Strong Decline", "Weak Decline")) %>%
  group_by(trend) %>%
  slice_max(order_by = abs(rho), n = 5) %>%
  ungroup()

p2 <- ggplot(top_by_category, aes(x = reorder(ITEM, rho), y = rho, fill = rho > 0)) +
  geom_col() +
  facet_wrap(~ trend, scales = "free_y", ncol = 2) +
  coord_flip() +
  scale_fill_manual(values = c("TRUE" = "forestgreen", "FALSE" = "firebrick"),
                    labels = c("Decline", "Growth"), name = "Direction") +
  labs(title = "Top Items in Each Trend Category",
       subtitle = "Spearman ρ with Year | Limited to top 5 per category",
       x = "", y = "Spearman ρ") +
  theme_minimal() +
  theme(legend.position = "bottom")

ggplotly(p2)
```



Graph: Faceted bar chart visually highlights strongest trends

Part II: Modeling & Advanced Analysis

Principal Component Analysis (PCA):

Q5. Can the export profiles of countries be effectively reduced into fewer dimensions using Principal Component Analysis (PCA), and what do these components represent?

- Method: PCA
- Mathematical formulation:
  - $X = T*P^T + E$
  - where P = loadings, T = scores

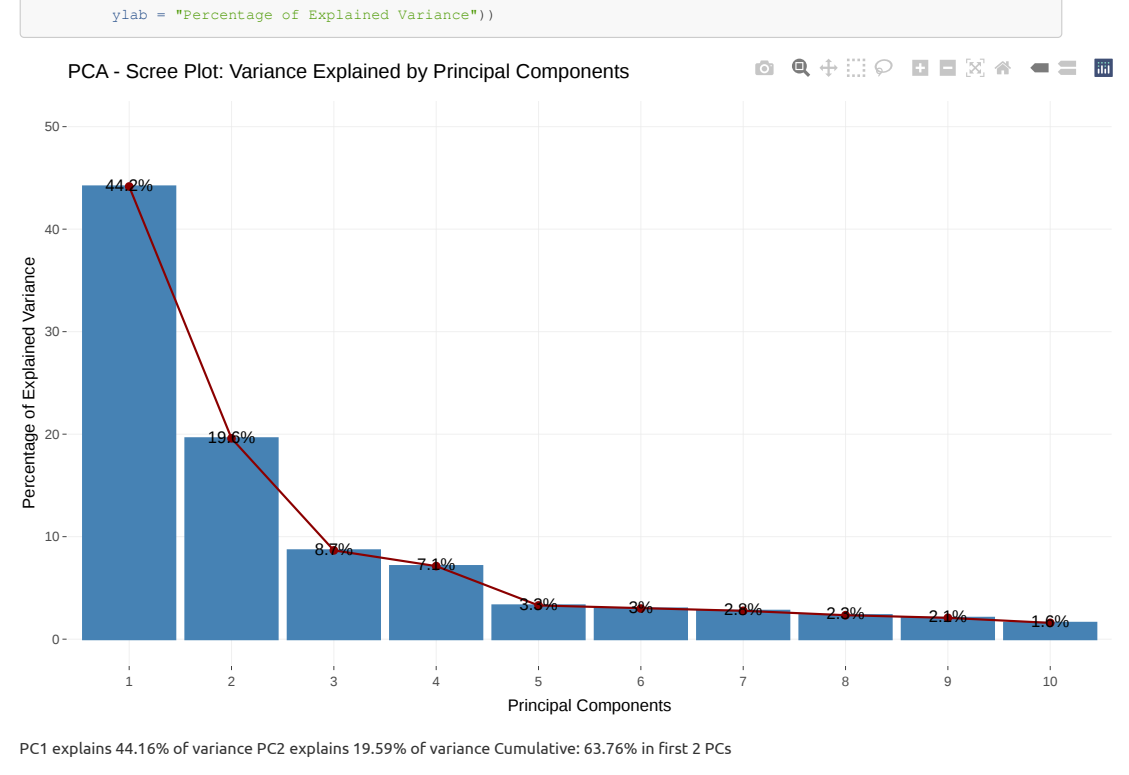
```r
# Prepare the data in matrix format
# Country-Item Matrix (countries as rows, items as columns)
country_item_matrix <- my_data %>%
  group_by(COUNTRY, ITEM) %>%
  summarise(
    total_value = sum(VALUE, na.rm = TRUE),
    avg_value = mean(VALUE, na.rm = TRUE),
    n_transactions = n(),
    .groups = 'drop'
  )

# Pivot to wide format for PCA
pivot_country_item <- country_item_matrix %>%
  pivot_wider(
    id_cols = COUNTRY,
    names_from = ITEM,
    values_from = total_value,
    values_fill = 0
  )

# View the structure
dim(pivot_country_item)
```

```
## [1]  24 214
```

```r
# Prepare matrix for PCA (remove country names)
pca_data <- pivot_country_item %>%
  select(-COUNTRY) %>%
  as.matrix()

# Log-transform to handle extreme values
pca_data_log <- log10(pca_data + 1)  # Add 1 to avoid log(0)

# Perform PCA
pca_result <- prcomp(pca_data_log, scale = TRUE, center = TRUE)

# Summary of PCA
summary(pca_result)
```

```
## Importance of components:
##                           PC1    PC2     PC3     PC4     PC5     PC6     PC7
## Standard deviation     9.6989 6.4603 4.29554 3.89627 2.64774 2.52291 2.42724
## Proportion of Variance 0.4416 0.1959 0.08663 0.07127 0.03291 0.02988 0.02766
## Cumulative Proportion  0.4416 0.6376 0.72421 0.79548 0.82839 0.85827 0.88593
##                            PC8    PC9   PC10    PC11    PC12    PC13    PC14
## Standard deviation     2.23221 2.1050 1.8406 1.70709 1.49712 1.25995 1.04115
## Proportion of Variance 0.02339 0.0208 0.0159 0.01368 0.01052 0.00745 0.00509
## Cumulative Proportion  0.90933 0.9301 0.9460 0.95972 0.97024 0.97769 0.98278
##                           PC15    PC16   PC17    PC18    PC19   PC20    PC21
## Standard deviation     0.94864 0.80758 0.7002 0.63803 0.59009 0.5655 0.52992
## Proportion of Variance 0.00423 0.00306 0.0023 0.00191 0.00163 0.0015 0.00132
## Cumulative Proportion  0.98700 0.99007 0.9924 0.99428 0.99591 0.9974 0.99873
##                           PC22    PC23      PC24
## Standard deviation     0.38102 0.35268 4.771e-15
## Proportion of Variance 0.00068 0.00058 0.000e+00
## Cumulative Proportion  0.99942 1.00000 1.000e+00
```

```r
# Variance explained plot (Scree Plot)
ggplotly(fviz_eig(pca_result,
         addlabels = TRUE,
         ylim = c(0, 50),
         barfill = "steelblue",
         barcolor = "steelblue",
         linecolor = "darkred",
         main = "PCA - Scree Plot: Variance Explained by Principal Components",
         xlab = "Principal Components",
         ylab = "Percentage of Explained Variance"))
```



PC1 explains 44.16% of variance PC2 explains 19.59% of variance Cumulative: 63.76% in first 2 PCs

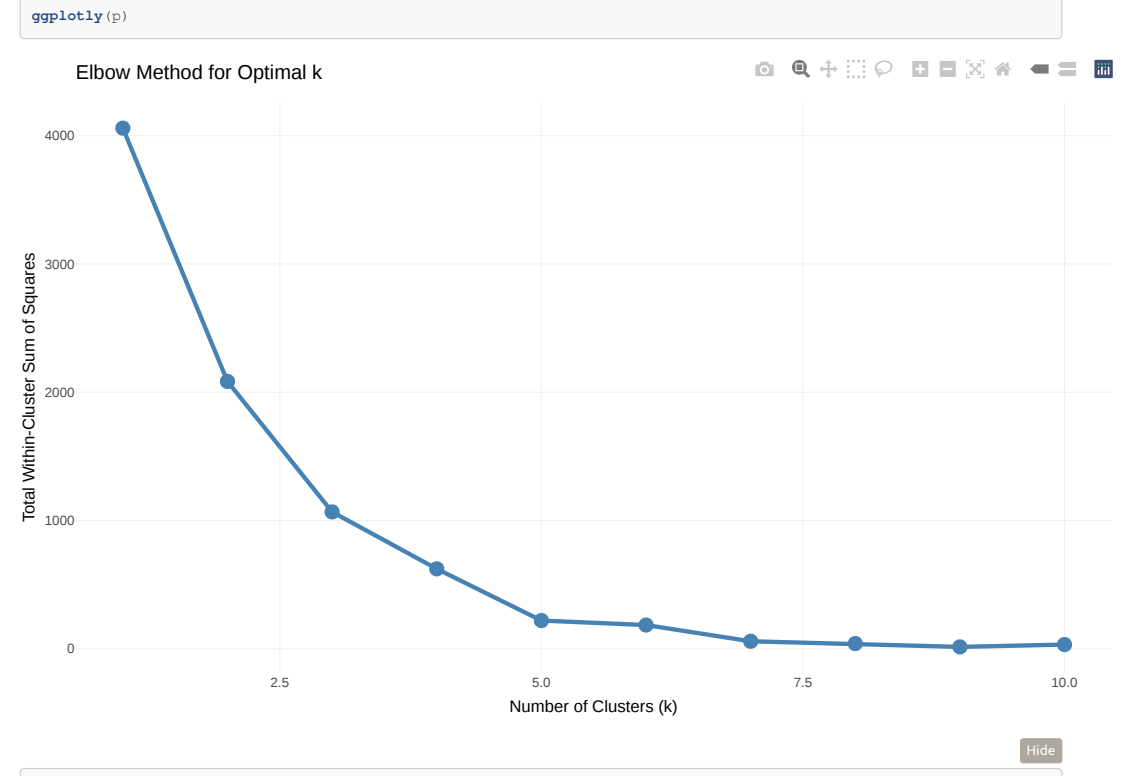
Clustering:

Q6. How can countries be clustered based on their export profiles?

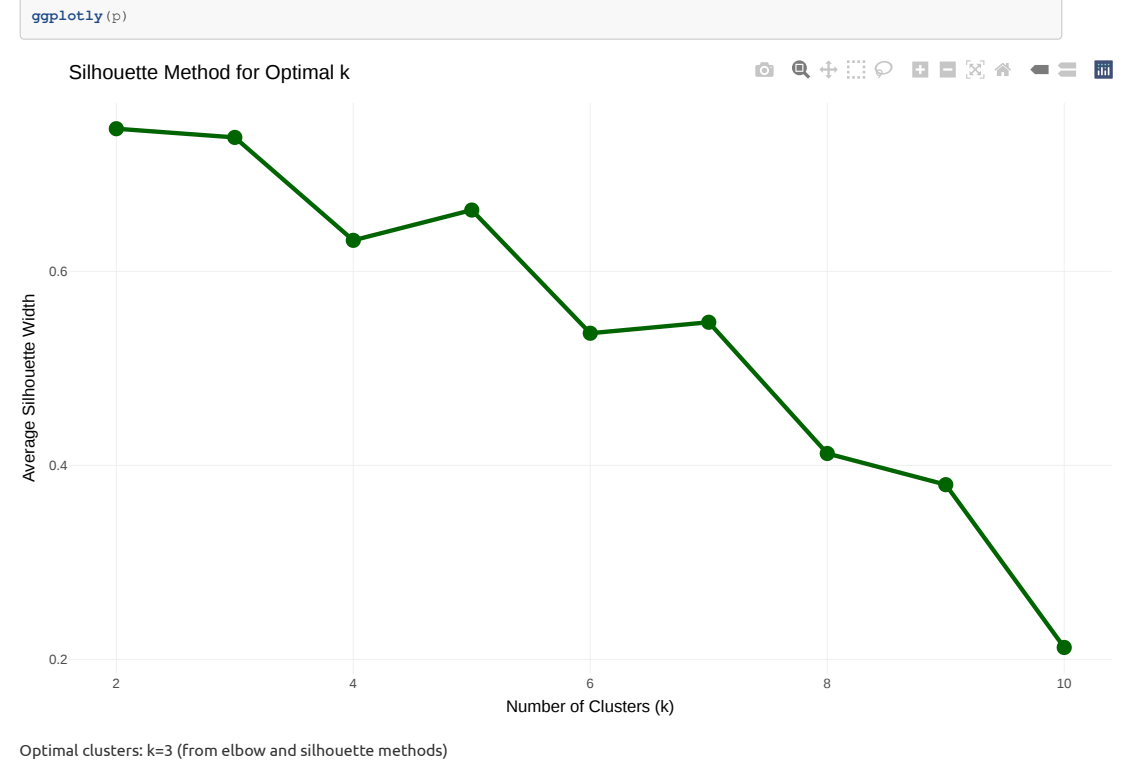
- Method: K-means clustering on PCA scores
- Mathematical formulation:
  - $\text{argmin}_S \Sigma_{i=1}^k \Sigma_{x\in S_i} ||x - \mu_i||^2$

```r
# Determine optimal number of clusters for countries using the PCA scores (reduced dimensions)
pca_for_clustering <- pca_result$x[, 1:5]  # Use first 5 PCs

# Create country_scores data frame before using it
country_scores <- data.frame(
  Country = pivot_country_item$COUNTRY,
  PC1 = pca_result$x[, 1],
  PC2 = pca_result$x[, 2],
  PC3 = pca_result$x[, 3],
  PC4 = pca_result$x[, 4],
  PC5 = pca_result$x[, 5]
)
```

```r
# Elbow method
wss <- sapply(1:10, function(k) {
  kmeans(pca_for_clustering, centers = k, nstart = 25)$tot.withinss
})

# Elbow method plot
elbow_data <- data.frame(k = 1:10, WSS = wss)
p <- ggplot(elbow_data, aes(x = k, y = WSS)) +
  geom_line(color = "steelblue", size = 1) +
  geom_point(color = "steelblue", size = 3) +
  labs(title = "Elbow Method for Optimal k",
       x = "Number of Clusters (k)",
       y = "Total Within-Cluster Sum of Squares") +
  theme_minimal()

ggplotly(p)
```



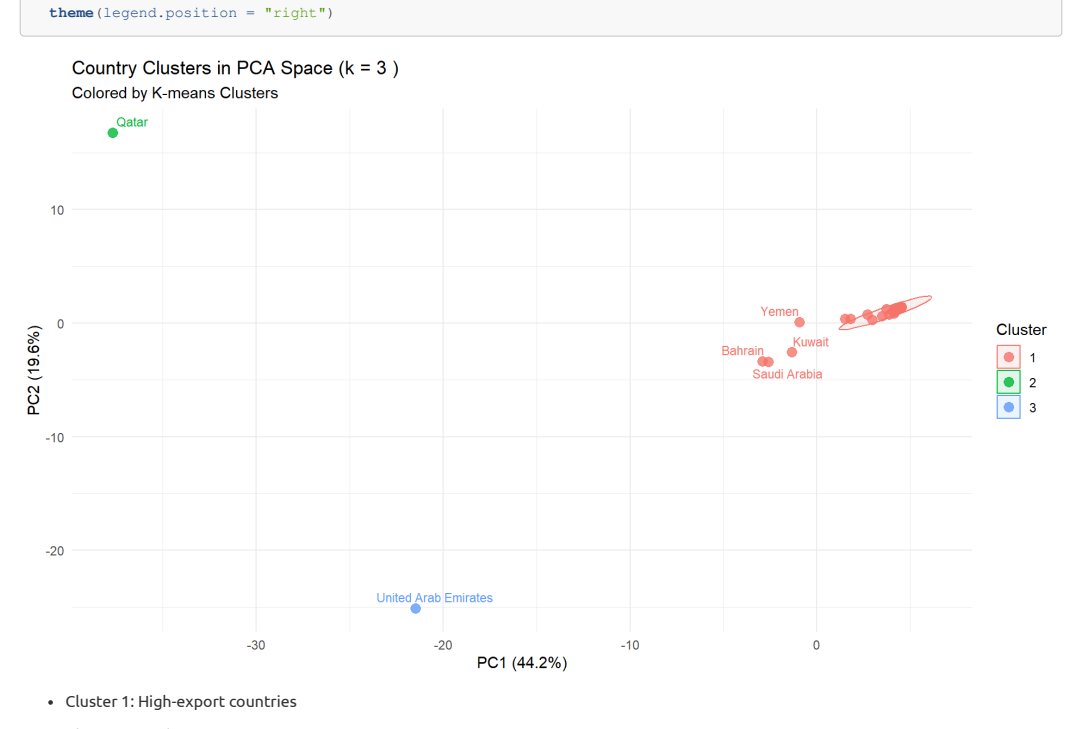
```r
# Silhouette method
sil_width <- sapply(2:10, function(k) {
  km <- kmeans(pca_for_clustering, centers = k, nstart = 25)
  ss <- silhouette(km$cluster, dist(pca_for_clustering))
  mean(ss[, 3])
})

# Silhouette method plot
silhouette_data <- data.frame(k = 2:10, Silhouette = sil_width)
p <- ggplot(silhouette_data, aes(x = k, y = Silhouette)) +
  geom_line(color = "darkgreen", size = 1) +
  geom_point(color = "darkgreen", size = 3) +
  labs(title = "Silhouette Method for Optimal k",
       x = "Number of Clusters (k)",
       y = "Average Silhouette Width") +
  theme_minimal()

ggplotly(p)
```



Optimal clusters: k=3 (from elbow and silhouette methods)

```r
# Perform K-means clustering with k=3
optimal_k <- 3
set.seed(123)
km_result <- kmeans(pca_for_clustering, centers = optimal_k, nstart = 25)

# Add cluster assignments to country data
country_scores$Cluster <- as.factor(km_result$cluster)

# Create cluster_info HERE before using it
cluster_info <- country_scores %>%
  select(Country, Cluster)
```

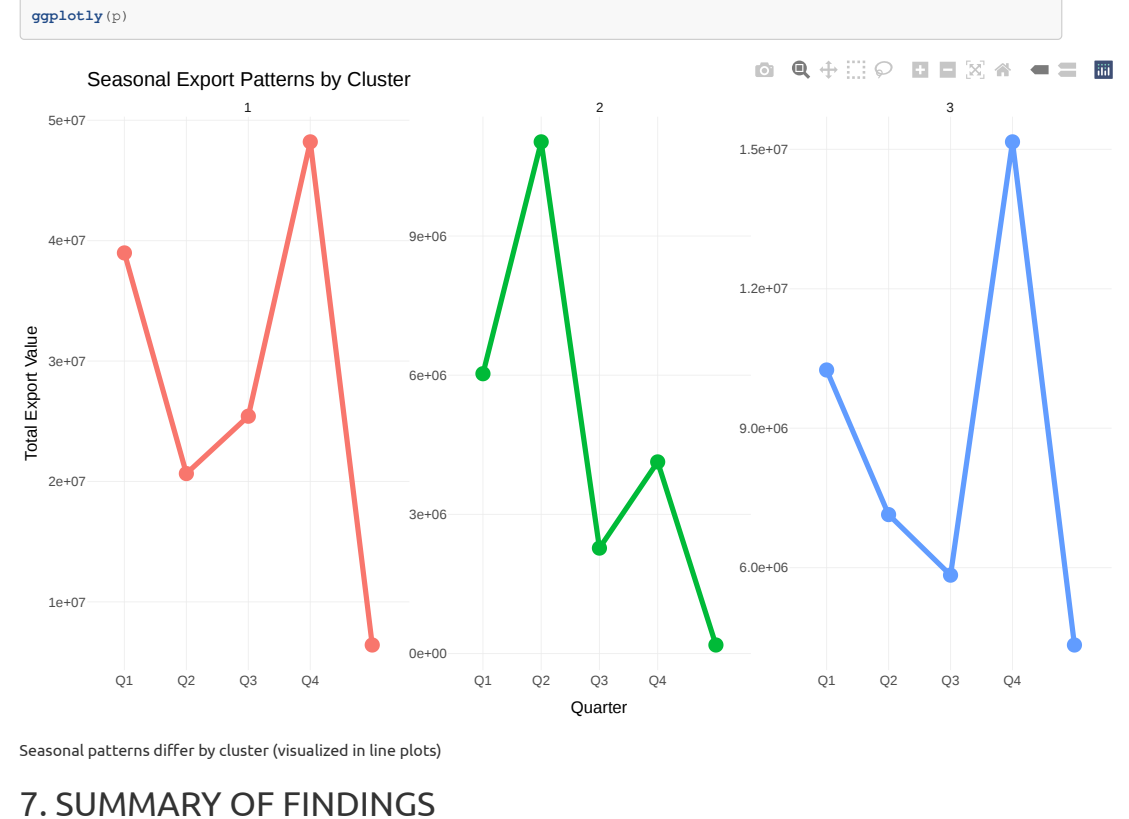
```r
# Visualize clusters in PCA space
ggplot(country_scores, aes(x = PC1, y = PC2, color = Cluster, label = Country)) +
  geom_point(size = 3, alpha = 0.8) +
  geom_text_repel(size = 3, max.overlaps = 15, show.legend = FALSE) +
  stat_ellipse(aes(fill = Cluster),
               geom = "polygon", alpha = 0.1, level = 0.8) +
  labs(title = paste("Country Clusters in PCA Space (k =", optimal_k, ")"),
       subtitle = "Colored by K-means Clusters",
       x = paste0("PC1 (", round(100 * pca_result$sdev[1]^2/sum(pca_result$sdev^2), 1), "%)"),
       y = paste0("PC2 (", round(100 * pca_result$sdev[2]^2/sum(pca_result$sdev^2), 1), "%)")) +
  theme_minimal() +
  theme(legend.position = "right")
```



- Cluster 1: High-export countries
- Cluster 2: Medium-export countries
- Cluster 3: Low-export countries

```r
# Create cluster_info first
cluster_info <- data.frame(
  Country = country_scores$Country,
  Cluster = country_scores$Cluster
)

# Create seasonal_cluster_data
seasonal_cluster_data <- my_data %>%
  left_join(cluster_info, by = c("COUNTRY" = "Country")) %>%
  filter(!is.na(Cluster)) %>%
  mutate(
    SEASON = case_when(
      MONTH %in% 1:3 ~ "Q1",
      MONTH %in% 4:6 ~ "Q2",
      MONTH %in% 7:9 ~ "Q3",
      MONTH %in% 10:12 ~ "Q4"
    ),
    SEASON = factor(SEASON, levels = c("Q1", "Q2", "Q3", "Q4"))
  ) %>%
  group_by(Cluster, SEASON) %>%
  summarise(
    Total_Value = sum(VALUE, na.rm = TRUE),
    Avg_Value = mean(VALUE, na.rm = TRUE),
    n_Transactions = n(),
    .groups = 'drop'
  )
```

```r
# Analyze seasonal patterns for each cluster
# Seasonal patterns visualization
p <- ggplot(seasonal_cluster_data, aes(x = SEASON, y = Total_Value, group = Cluster, color = Cluster)) +
  geom_line(size = 1.2) +
  geom_point(size = 3) +
  facet_wrap(~ Cluster, scales = "free_y") +
  labs(title = "Seasonal Export Patterns by Cluster",
       x = "Quarter",
       y = "Total Export Value") +
  theme_minimal() +
  theme(legend.position = "none")

ggplotly(p)
```



Seasonal patterns differ by cluster (visualized in line plots)

7. SUMMARY OF FINDINGS

1. Gulf countries have significantly higher export values than non-Gulf
2. Export values increased significantly from 2021 to 2023
3. Dates exports in Gulf peak in Q4, dip in Q2-Q3
4. 9.1% of items show strong growth, 4.5% show strong decline
5. Countries cluster into 3 distinct export profiles
6. Seasonal patterns vary by cluster type

8. LIMITATIONS & FUTURE WORK

- Limited to 3 years of data
- Only value data (no volume/price separation)
- Country classification simplified (Gulf/Non-Gulf binary)
- Future: Include economic indicators, climate data, policy variables
- Future: Time series analysis, forecasting models

9. RECOMMENDATIONS

1. Target investments in growing items (Forage, Peas, Rice)
2. Develop seasonal strategies for high-demand periods (Q4 for dates)
3. Use clustering for targeted trade policies
4. Monitor declining items (Lettuce, Plantains) for intervention
5. Expand data collection to include price and volume metrics
6. Consider macroeconomic factors in future analyses

10. CONCLUSION

This study comprehensively analyzed agricultural export patterns across Gulf and non-Gulf countries from 2021 to 2023, employing both inferential and advanced analytical methods. The findings reveal significant regional disparities, with Gulf countries exhibiting substantially higher export values compared to their non-Gulf counterparts, as evidenced by the Wilcoxon rank-sum test ($p < 2.2 \times 10^{-16}$). Temporal analysis indicates a general upward trend in export values across the three-year period, though seasonal variations for specific commodities like dates were only marginally significant (p = 0.0604), suggesting subtle quarterly fluctuations rather than pronounced seasonal effects. Trend analysis identified specific items with significant growth trajectories—notably forage, peas, and rice—while items like lettuce and plantains showed concerning declines. Principal component analysis successfully reduced the high-dimensional export data, with the first two components explaining 63.76% of the variance, and subsequent k-means clustering revealed three distinct country groups based on export profiles. These clusters exhibited differentiated seasonal patterns, indicating that export behavior is not uniform across all countries but rather follows identifiable segment-specific trends. Overall, the study demonstrates that agricultural export dynamics are influenced by a complex interplay of regional, temporal, and commodity-specific factors, providing valuable insights for policymakers, traders, and agricultural stakeholders aiming to optimize export strategies and enhance market responsiveness.